Phase 4 — Paper Figures Spec

2026-04-20 · docs/reports/2026-04-19-phase4-eigenvalue-study/figures-spec.typ

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Target paper: `_NeurIPS_2026__Understanding_Graph_Denoising.pdf`. Three figures (F1–F3) summarise the Phase 4 improvement-gap surrogate sweep. All figures read the same CSV and share one style module so the house look stays consistent.

Goal

For a given graph dataset and noise level ε, quantify the extent to which the noisy top-k eigenvalues $\tilde{\Lambda}_k$ determine the projection of the clean adjacency $B = \hat{V}_k^T A \hat{V}_k$ into the noisy top-k eigenbasis. Equation (18) of the target draft defines the improvement gap

$$\ell_{\text{lin}} - \ell_f = \mathbb{E}\left\|\mathbb{E}\left[B \mid \tilde{\Lambda}_k\right] - \mathbb{E}[B]\right\|_F^2,$$

i.e. the between-graph variance of the Bayes-optimal predictor of B given $\tilde{\Lambda}_k$. The figures report a bias-corrected finite-sample estimate of this population quantity, normalised by $\operatorname{tr}\operatorname{Cov}(B)$, so that a reader can (i) read the fraction of B's variance that $\tilde{\Lambda}_k$ determines on each dataset, (ii) compare datasets on a common $[0, 1]$ scale, and (iii) verify that the estimate rises with known community strength (synthetic SBM diversity) and reflects the community content of real benchmarks. Making this quantity legible, bias-corrected, and reproducible across k, ε, noise model, and frame convention is the purpose of the figure set.

Shared conventions

- **Data source**: `docs/reports/2026-04-19-phase4-eigenvalue-study/phase4_sweep.csv` (25 600 rows).
- **Headline cell** (unless stated otherwise): `frame_mode='frechet'`, `estimator_label='knn_top_k'`, `noise_type='gaussian'`. This is the cell whose margins the README headline table reports.
- **Script**: `scripts/plot_phase4_figures.py`, uv-style `# /// script` header with `matplotlib>=3.8`, `pandas>=2.0`, `numpy>=1.24`. No side effects outside the figures directory.
- **Output dir**: `docs/reports/2026-04-19-phase4-eigenvalue-study/figures/`. Emit **both** `.pdf` (for LaTeX inclusion) and `.png` at 300 dpi (for quick review and Slack).
- **Sizes**: NeurIPS single-column 3.25″, double-column 6.75″. F1 and F3 are double-column; F2 is single-column.
- **Style**: `plt.rcParams['font.family'] = 'serif'`, `font.size=8`, `axes.labelsize=8`, `legend.fontsize=7`. No title on the figure itself — caption lives in the `.tex`. Panel titles OK when there are multiple subplots. Tight layout; `bbox_inches='tight'` on save.
- **Error semantics**: error bars and ribbons are ± 1 SD across the five seeds $\{42, 123, 2024, 7, 11\}$. Points are seed means.
- **Diversity knob** (definition). `diversity` $\in [0, 1]$ scales the per-graph sampling-range width of each tuple-ranged SBM hyperparameter toward its midpoint: for a range (l, h) with midpoint $\mu = (l + h)/2$ and half-width $w = (h - l)/2$, the effective sampling range is $(\mu - wd, \mu + wd)$. At $d = 0$ every graph is drawn from μ (homogeneous batch); at $d = 1$ the full range is used (maximally heterogeneous). Phase 4 fixes $n = 50$ nodes, 4 blocks, and uses $p_{\text{intra}} \in (0.3, 0.9)$, $p_{\text{inter}} \in (0.01, 0.2)$. See F0 below for visual examples.
- **Palette**: synthetic SBMs use a viridis gradient over diversity $\{0.0, 0.33, 0.67, 1.0\}$ so monotonicity reads at a glance. Real datasets use tab10 warm colours (`spectre_sbm` red, `enzymes` orange, `proteins` green, `collab` purple). Null ribbon is neutral grey `#b0b0b0` at alpha 0.35. Pass / fail bar colouring in F3: pass `#3b7dd8`, fail `#b0b0b0`.

Metric definition — fraction of variance explained (FVE)

Let $B_i = \hat{V}_{k,i}^T A_i \hat{V}_{k,i} \in \mathbb{R}^{k\times k}$ be the clean adjacency of graph i re-expressed in the noisy top-k eigenbasis, and let $\tilde{\Lambda}_{k,i} \in \mathbb{R}^k$ be the corresponding noisy eigenvalues (both in the dataset-wide Fréchet frame; see `figures.md` for why). The metric on all figure y-axes derives from

$$\mathrm{FVE}(k) = \frac{\mathrm{Var}\Big(\mathbb{E}\big[B_i \mid \tilde{\Lambda}_{k,i}\big]\Big)}{\mathrm{Var}(B_i)} \in [0,1].$$

Both variances are Frobenius-sum variances taken across the graph population. By the law of total variance this is exactly the **coefficient of determination (R^2) of the Bayes-optimal predictor** of B from $\tilde{\Lambda}_k$: it measures the fraction of B's Frobenius variance that **any** function of $\tilde{\Lambda}_k$ could predict. $\mathrm{FVE} = 0$ means $\tilde{\Lambda}_k$ carries no information about B; $\mathrm{FVE} = 1$ means $\tilde{\Lambda}_k$ determines B entirely.

The **calibrated FVE margin** $\mathrm{FVE}_{\mathrm{real}} - \mathrm{FVE}_{\mathrm{null}}$ is the quantity F2 and F3 plot directly, where $\mathrm{FVE}_{\mathrm{null}}$ is FVE recomputed with $\tilde{\Lambda}_k$ permuted across graphs (so feature–target pairing is destroyed while marginals are preserved).

Why FVE rather than alternatives

Alternative	Why we didn't pick it
Raw $\hat{g}_k$ (unnormalised numerator)	Has units of Frobenius² and scales mechanically with $\mathrm{trace}\ \mathrm{Cov}(B)$, which itself grows with edge density and n. Cross-dataset and cross-k comparisons become meaningless. FVE normalises this away.
Linear R^2 (OLS regression of $\mathrm{vec}(B)$ on $\tilde{\Lambda}_k$)	Assumes a linear relationship; the improvement-gap theory (eq. 18) is explicitly about the **non-linear** Bayes gap $\ell_{\mathrm{lin}} - \ell_f$. Using linear R^2 would collapse the quantity we care about to zero by construction.
Mutual information $I\big(B; \tilde{\Lambda}_k\big)$	Correct in principle but very hard to estimate for matrix-valued B at our sample sizes ($N \approx 68$–3700). Known estimators (kNN-MI of Kraskov, KSG) have larger and less-predictable finite-sample bias than the conditional-mean estimator, and no natural $[0,1]$ scale.
HSIC / dCor (kernel / distance dependence measures)	Detect arbitrary dependence, but lose the variance-share interpretation. A HSIC value of 0.3 does not translate to "30% of the variance in B is recoverable from $\tilde{\Lambda}_k$" — it just says the two are statistically dependent. Reviewers will ask "how much is that", and we would have no answer.
MSE of a fitted regressor (train/test MLP, GP, kernel ridge)	Depends on model class choice and hyperparameter tuning. Each choice raises a "did you pick the best class?" objection. The Bayes-optimal benchmark underlying FVE sidesteps this entirely.
Spearman / Pearson on scalar summaries (e.g. spectral gap vs. $\mathrm{trace}(B)$)	Too coarse: B is a $k \times k$ matrix, and projecting it to a scalar before computing a rank correlation discards the information the paper's surrogate explicitly keeps.

Estimator definition — kNN conditional mean

We estimate $\mathbb{E}\left[B_i \mid \tilde{\Lambda}_{k,i}\right]$ by nearest-neighbour averaging in $\tilde{\Lambda}_k$-space:

1. Compute the Euclidean pairwise distance matrix $D \in \mathbb{R}^{N\times N}$ with $D_{ij} = \left\|\tilde{\Lambda}_{k,i} - \tilde{\Lambda}_{k,j}\right\|_2$.
2. For each graph i, let $\mathcal{N}_i$ be the indices of the $\boldsymbol{m = 10}$ nearest neighbours (excluding i itself — the self-entry sits at distance 0).
3. Form the kNN conditional-mean estimate $\hat{B}_i = \left(\frac{1}{m}\right)\sum_{j\in\mathcal{N}_i} B_j$.
4. Compute $\hat{g}_k = \left(\frac{1}{N}\right)\sum_i \left\|\hat{B}_i - \mu_B\right\|_F^2$ where $\mu_B = \left(\frac{1}{N}\right)\sum_i B_i$.
5. Report $\mathrm{FVE} = \hat{g}_k / \left(\frac{1}{N}\right)\sum_i \|B_i - \mu_B\|_F^2$.

The CSV's `estimator_label='knn_top_k'` column corresponds to this choice with $\tilde{\Lambda}_k$ as the full top-k eigenvalue vector. The other labels (`knn_1d`, `bin_1d`, `invariants_knn`) are sensitivity variants and do not feed the main paper figures.

Why kNN with $m = 10$

- **Nonparametric.** No distributional assumption on $\left(B, \tilde{\Lambda}_k\right)$. We don't know, a priori, the functional form of $\mathbb{E}\left[B \mid \tilde{\Lambda}_k\right]$ — the whole premise of the paper is that the relationship is nonlinear in a specific way — so a nonparametric regressor is the right default.
- **No bandwidth selection.** m is the only hyperparameter. Kernel regression would require choosing a kernel and a bandwidth (Silverman's rule, cross-validation, Lepski); each choice is another reviewer hook. Changing m in $\{5, 10, 20\}$ does not change the qualitative picture in our sensitivity runs.
- **Predictable finite-sample bias.** The average over m independent neighbours introduces a known $\approx 1/m$ residual variance under H_0 (features independent of targets), which appears as a flat ≈ 0.10 floor in the permutation null. This floor is what the calibrated margin subtracts out. No other estimator we considered has such a cleanly-measurable bias; MLPs and kernel regressors mix bias and variance in harder-to-isolate ways.
- **Consistent.** $\hat{g}_k \to g_k$ as $N \to \infty$, $m \to \infty$, $m/N \to 0$ (standard kNN-regression consistency). At our sample sizes ($N \geq 68$) $m = 10$ sits comfortably in the safe regime.
- **Computationally trivial.** One $N \times N$ distance matrix plus a single stack-and-average per graph. Every cell in the 25 600-row sweep runs in milliseconds.

Calibrated margin as a lower bound on the Bayes FVE

Decompose the kNN predictor as $\hat{m}(\lambda) = m^*(\lambda) + \mathrm{bias}(\lambda) + \xi(\lambda)$, where $m^*(\lambda) = \mathbb{E}\left[B \mid \tilde{\Lambda}_k = \lambda\right]$ is the Bayes-optimal predictor, $\mathrm{bias}(\lambda) = \mathbb{E}[\hat{m}(\lambda)] - m^*(\lambda)$ is the smoothing error from averaging over m neighbours whose $\tilde{\Lambda}$ is not exactly λ, and $\xi(\lambda)$ is the residual noise of the neighbour average around $\mathbb{E}[\hat{m}(\lambda)]$. Then

$$\mathrm{Var}\left(\hat{m}\left(\tilde{\Lambda}_k\right)\right) \approx \underbrace{\mathrm{Var}\left(m^*\left(\tilde{\Lambda}_k\right)\right)}_{\text{true Bayes signal}} + \underbrace{O(h^2)}_{\text{smoothing attenuation}} + \underbrace{\sigma^2/m}_{\text{neighbour-averaging inflation}},$$

with effective bandwidth $h \sim (m/N)^{1/k}$ for k-dimensional $\tilde{\Lambda}_k$. This is the standard kNN-regression bias-variance decomposition under Lipschitz m^*: pointwise bias is $O(h)$ so $\mathrm{bias}^2 = O(h^2) = O\left((m/N)^{2/k}\right)$, and neighbour-averaging variance is $O(\sigma^2/m)$ (Györfi, Kohler, Krzyżak & Walk 2002, ch. 6 on kNN estimates; Biau & Devroye 2015, Part II ch. 14 "Rates of Convergence"; Ayano 2012, which further shows that for (p, C)-smooth m^* plain kNN achieves the minimax rate only up to $p = \frac{3}{2}$, so stronger-smoothness regimes — e.g. twice-differentiable m^* giving the faster $O(h^4)$ bias² — are unreachable without local-linear corrections we do not apply).

The inflation term σ^2/m is approximately invariant under permutation (same m, N, and marginal noise), so subtracting $\mathrm{FVE}_{\mathrm{null}}$ removes it cleanly. The attenuation term, however,

is absent from the null: under H_0 the true m^* is constant and there is nothing to over-smooth. It therefore survives the subtraction as a **negative** contribution to $\mathrm{FVE}_{\mathrm{real}} - \mathrm{FVE}_{\mathrm{null}}$. The calibrated margin we report is, in consequence, a **conservative lower bound** on the true Bayes FVE — the population quantity from eq. (18):

$$\mathrm{FVE}_{\mathrm{real}} - \mathrm{FVE}_{\mathrm{null}} \leq \mathrm{FVE}_{\mathrm{Bayes}} + O\big((m/N)^{2/k}\big).$$

Numerically the attenuation bound $(m/N)^{2/k}$ at $m = 10$ and our sample sizes is:

k	N = 200	N = 3700
4	0.22	0.05
8	0.47	0.23
16	0.69	0.48
32	0.83	0.69

At $k = 4$ with large N the bound is tight. At $k = 16$ and $k = 32$ it becomes loose, and our calibrated margin increasingly understates the true Bayes R^2. Two practical consequences: (i) a *positive* calibrated margin at high k is still evidence for real signal, because the estimator errs conservatively; (ii) a *shrinking* margin as k grows in F1 / F2 cannot be attributed solely to loss of signal — part of it is estimator conservatism. We flag this in the figure captions rather than correcting it in post, because the bias constant depends on unknown smoothness properties of m^* and on the intrinsic (rather than ambient) dimension of $\tilde{\Lambda}_k$, which for eigenvalues of structured adjacency matrices is often smaller than k.

How to read the figures

Two interpretive handles the paper will lean on:

1. **A nonzero gap between the real curve and the null ribbon implies improvability, period.** The gap is a conservative lower bound on the population improvement gap $\ell_{\mathrm{lin}} - \ell_f$, so positivity of the gap — once it exceeds seed-level noise, which is what the 0.10 pass threshold operationalises — guarantees that some non-constant function of $\tilde{\Lambda}_k$ genuinely beats the marginal mean as a predictor of B. The second clause of the pass criterion ($\mathrm{FVE}_{\mathrm{null}} < 0.30$) guards against small-$N$ regimes where bias inflation leaves no headroom.
2. **The decay of the curve as ε grows measures noise sensitivity of the exploitable signal.** At $\varepsilon = 0$ the noisy and clean eigenframes coincide, B is deterministic in Λ_k, and FVE saturates near 1 (modulo smoothing bias). As ε increases, $\tilde{\Lambda}_k$ drifts from Λ_k, $\mathbb{E}\big[B \mid \tilde{\Lambda}_k\big]$ spreads, and FVE falls. The shape of the decay is the empirical answer to "how robust is the dataset's spectral signal to additive noise?" — slow decay marks resilient, low-frequency structure; fast decay marks fragile, high-frequency structure that noise washes out first.

References (validated)

- Györfi L., Kohler M., Krzyżak A. & Walk H. (2002). *A Distribution-Free Theory of Nonparametric Regression.* Springer. DOI 10.1007/b97848. Chapter 6 gives the classical Lipschitz rate $O\big((k/n)^{2/d}\big) + O(1/k)$ for the kNN regression estimate.
- Biau G. & Devroye L. (2015). *Lectures on the Nearest Neighbor Method.* Springer Series in the Data Sciences. DOI 10.1007/978-3-319-25388-6. Part II ch. 14 ("Rates of Convergence") reproves and extends the kNN regression rates; ch. 8 introduces the estimator.

- Ayano T. (2012). *Rates of convergence for the k-nearest neighbor estimators with smoother regression functions.* Statistics & Probability Letters. ScienceDirect. Shows plain kNN attains $n^{-2p/(2p+d)}$ only for $p \in \left(0, \frac{3}{2}\right]$; higher-order smoothness requires bias-reducing variants.

The permutation null as a bias diagnostic, not a significance test

Shuffling $\tilde{\Lambda}_k$ across graphs preserves the marginal distribution of the features but breaks the pairing with B, so under the null the true conditional mean is constant: $\mathbb{E}\left[B \mid \tilde{\Lambda}_k^{\text{perm}}\right] \equiv \mu_B$. Any non-zero FVE_{null} is therefore estimator bias, not genuine structure. We report it and subtract it; we do **not** report a p-value. This matches the conceptual framing in the paper — the surrogate is an effect-size measurement of the Bayes gap, not a hypothesis test.

Pass criterion

A dataset cell passes when $\text{FVE}_{\text{real}} - \text{FVE}_{\text{null}} \geq 0.10$ **and** $\text{FVE}_{\text{null}} < 0.30$. The first clause is the signal requirement; the second clause guards against small-N regimes where the kNN bias inflates and leaves little headroom between the null and the real curve. The two thresholds were fixed before Phase 4 data collection and are the same thresholds the Phase 3 diversity-knob validation used.

F1 — FVE vs ε per dataset, Gaussian vs DiGress noise side-by-side

Question: does the FVE stay above the permutation null across ε, does it behave similarly across k, and does the picture change under the structured (DiGress) noise the paper's generative models actually use?

- **Layout**: 4 rows $\times$ 4 cols grid, 16 panels, shared y-axis per row and shared x-axis per column. Landscape, ≈ 6.75 times 6.0.
- **Row layout**: rows 1–2 are `noise_type='gaussian'` (top half); rows 3–4 are `noise_type='digress'` (bottom half). A thin horizontal rule between row 2 and row 3 plus a left-margin row-group label ("Gaussian" / "DiGress") makes the split visually explicit.
- **Column layout**: same dataset ordering in every row, one dataset per column — row-major within each half, top-left to bottom-right: `sbm_d0.00`, `sbm_d0.33`, `sbm_d0.67`, `sbm_d1.00` on the upper row of each half; `spectre_sbm`, `enzymes`, `proteins`, `collab` on the lower row of each half. Holding the column order constant across the Gaussian and DiGress halves makes vertical scanning within a dataset the natural comparison.
- **Per panel**: x-axis is `noise_level` $\in \{0.01, 0.05, 0.1, 0.15, 0.2\}$ linear. y-axis is FVE on range $[0, 0.8]$ (shared across all 16 panels). Foreground: four lines for $k \in \{4, 8, 16, 32\}$ over `permuted=False`, coloured by k with a sequential `plasma` cmap, markers + line + ± 1 SD error bars across seeds. Background ribbon: for each k, `fill_between` ± 1 SD of the `permuted=True` curve in the same k-colour at alpha 0.15 — keeps the k-specific null visible rather than collapsing to a single band. Dashed horizontal line at $y = 0.10$ (pass threshold). Panel subtitle = dataset name + retained N, e.g. `sbm_d0.67 (N=200)`; noise type is indicated by the row-group label, not repeated per panel.
- **Legend**: single legend bottom-centre with $k = \frac{4}{8}{16}{32}$ swatches plus a grey ribbon swatch labelled "permutation null (± 1 SD)".
- **Filename**: `F1_fve_vs_epsilon.pdf` / `.png`.

F2 — calibrated FVE margin vs diversity (synthetic SBM only)

Question: does the FVE track community strength monotonically, and does the effect size grow with k?

- **Layout**: single panel, single-column, ≈ 3.25 times 2.5.
- **Data**: subset to `dataset` $\in$ `{sbm_d0.00, sbm_d0.33, sbm_d0.67, sbm_d1.00}`, `noise_level=0.1`, headline cell (Fréchet, `knn_top_k`, Gaussian).
- **x-axis**: diversity $\in \{0.0, 0.33, 0.67, 1.0\}$ on a linear scale, ticks at the four grid points.
- **y-axis**: calibrated FVE margin $= \mathrm{FVE}_{\mathrm{real}} - \mathrm{FVE}_{\mathrm{null}}$ per seed, averaged across seeds (pairing at the seed level so error bars propagate seed-to-seed).
- **Lines**: one line per $k \in \{4, 8, 16, 32\}$, coloured with the same plasma cmap as F1. Markers at each diversity point; error bars = SD of the per-seed calibrated margin.
- **Dashed horizontal** at $y = 0.10$: pass threshold.
- **Legend**: upper-left inside the axes, 4 entries.
- **Filename**: `F2_calibrated_margin_vs_diversity.pdf` / `.png`.

F3 — dataset ranking at the headline cell

Question: which datasets carry a community signal the surrogate recovers, ordered by margin?

- **Layout**: single panel, double-column width (for readability), ≈ 6.75 times 2.75.
- **Data**: subset to headline cell + $k = 8$, `noise_level=0.1`, `noise_type='gaussian'`, `frame_mode='frechet'`, `estimator_label='knn_top_k'`. One row per dataset (seed-averaged), sorted by calibrated margin descending.
- **Chart**: horizontal bar chart. Bar length = calibrated FVE margin ($\mathrm{FVE}_{\mathrm{real}} - \mathrm{FVE}_{\mathrm{null}}$). Bar colour is blue if margin ≥ 0.10 **and** $\mathrm{FVE}_{\mathrm{null}} < 0.30$ (= pass per the README criterion), else grey. Black horizontal error bar on each bar: $\sqrt{\mathrm{Var}(\mathrm{FVE}_{\mathrm{real}})/5 + \mathrm{Var}(\mathrm{FVE}_{\mathrm{null}})/5}$ across seeds (standard-error of the difference of two means). Annotate each bar on the right with the numeric margin to two decimals. Vertical dashed line at $x = 0.10$ (pass threshold).
- **Axes**: y = dataset label; x = calibrated FVE margin (0 to ≈ 0.7).
- **Filename**: `F3_dataset_ranking.pdf` / `.png`.

Supplementary (rendered by default)

The same script also emits, without requiring any flag:

- **FS1 — per-graph frame ranking**: identical to F3 but with `frame_mode='per_graph'`. Demonstrates the Fréchet-frame contribution to the signal. Filename `FS1_dataset_ranking_per_graph.pdf` / `.png`.

The main paper cites F1–F3; FS1 goes to the appendix. (The former supplementary "F1-digress" is no longer needed: DiGress now appears as the bottom half of F1.)

Implementation notes

- Load CSV once, filter per figure. No reliance on row ordering — always group by the full `(dataset, noise_type, noise_level, frame_mode, estimator_label, permuted, k)` key before aggregating across seeds.
- Seed count assertion: for every cell used in the figures, assert `len(group) == 5`; fail loudly if not.
- Save each figure twice (PDF, PNG). The PDF vector output is what LaTeX will `\includegraphics`.
- No `plt.show()`. No interactive state leaking between figures — call `plt.close(fig)` after saving.
- The script prints a one-line manifest per figure: `wrote F1 (panels=16, series=4, points=320) → figures/F1_fve_vs_epsilon.{pdf,png}`.

Resolved design decisions

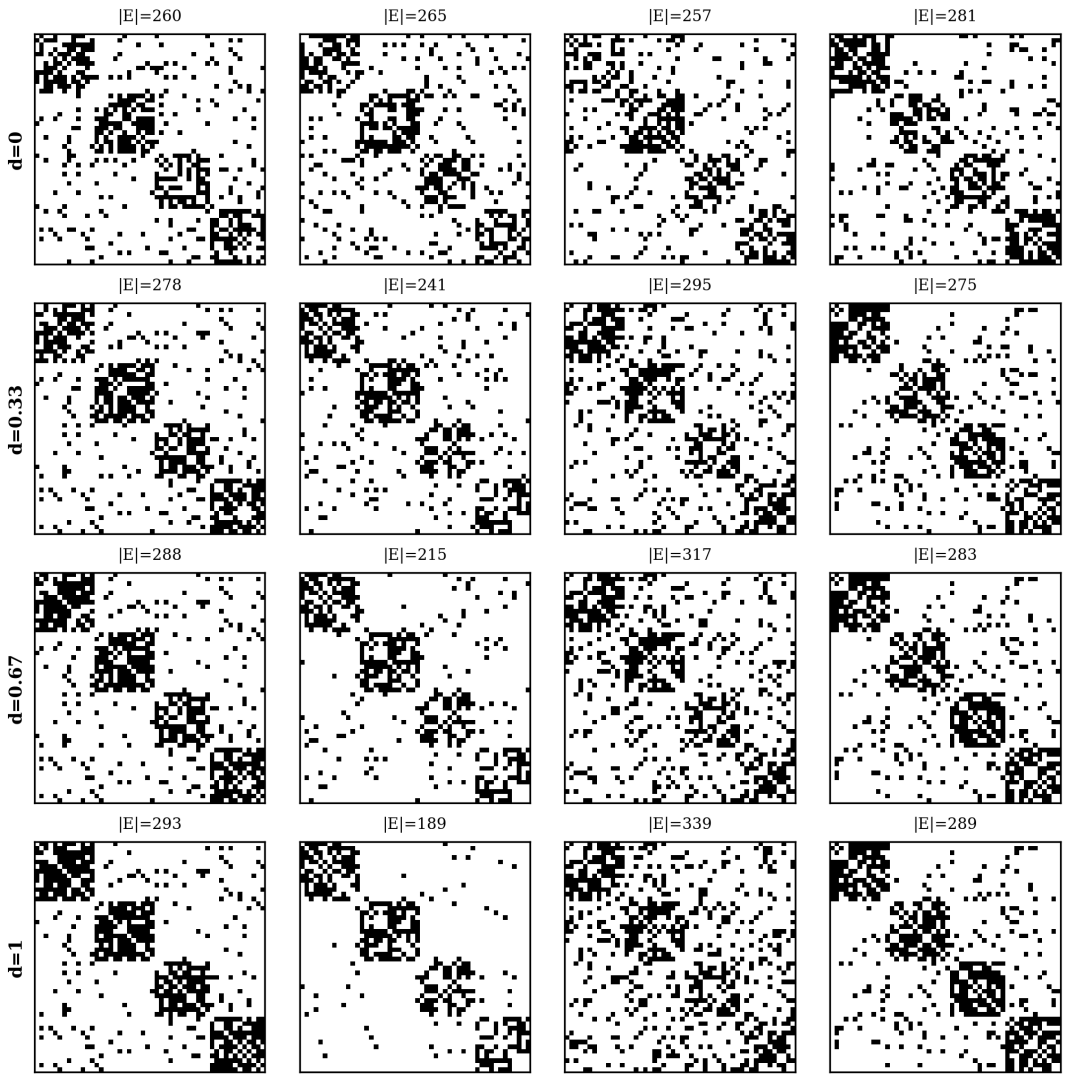
Captured here so the rationale stays with the spec even after the open-questions block is removed.

1. **F1 subplot grid**: 4 rows $\times$ 4 cols (was 2$\times$4 Gaussian-only). Gaussian on top half, DiGress on bottom, double-column width.
2. **Pass-threshold line on F1**: absolute $y = 0.10$, matching the README headline pass criterion. Not per-panel null-shifted.
3. **F2 k-lines**: all four $k \in \{4, 8, 16, 32\}$ rendered.
4. **Supplementary figures**: rendered by default — no `--supplementary` flag needed.
5. **DiGress in F1**: promoted to the bottom half of F1 (side-by-side with Gaussian) rather than relegated to supplementary. Former "F1-digress" supplementary is subsumed.

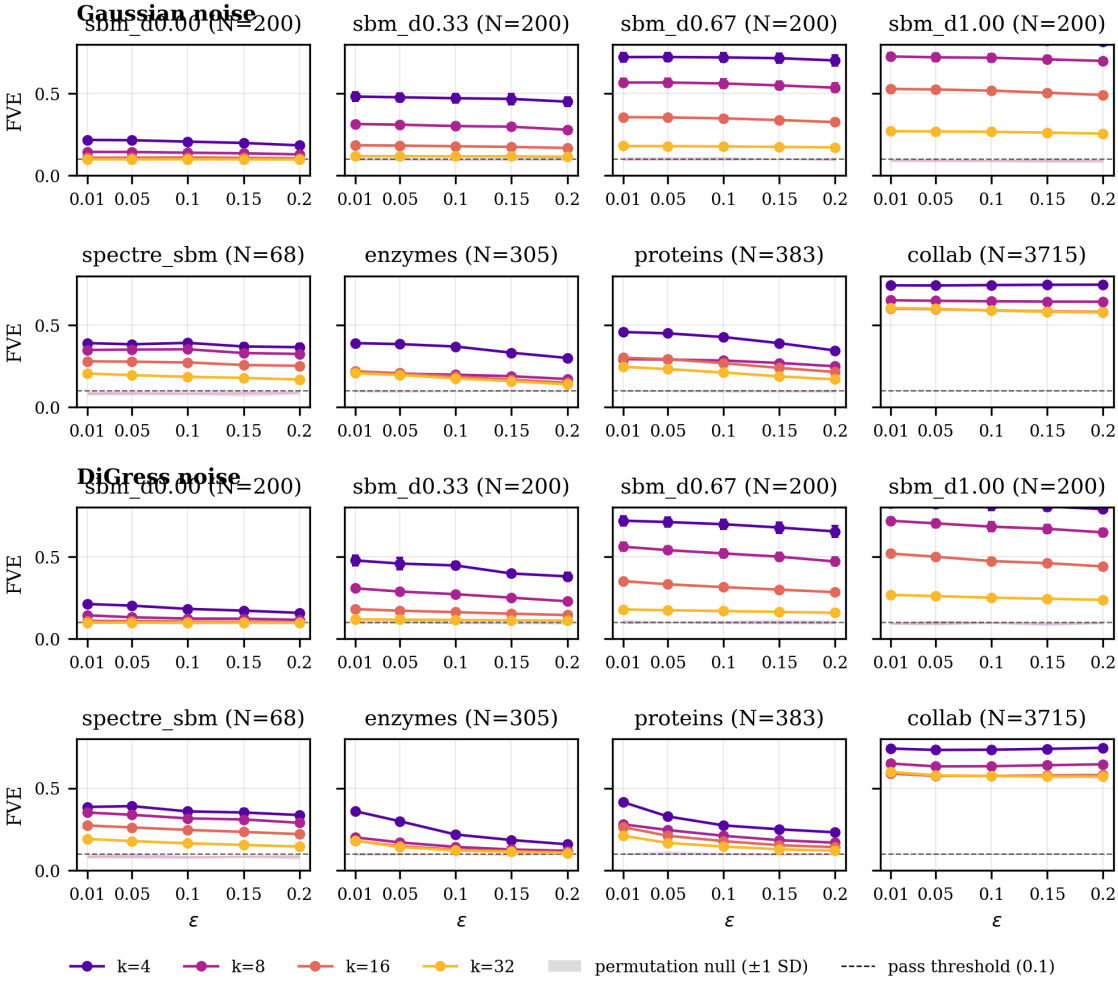
Generated figures

F0 — SBM-diversity knob, illustrative graphs

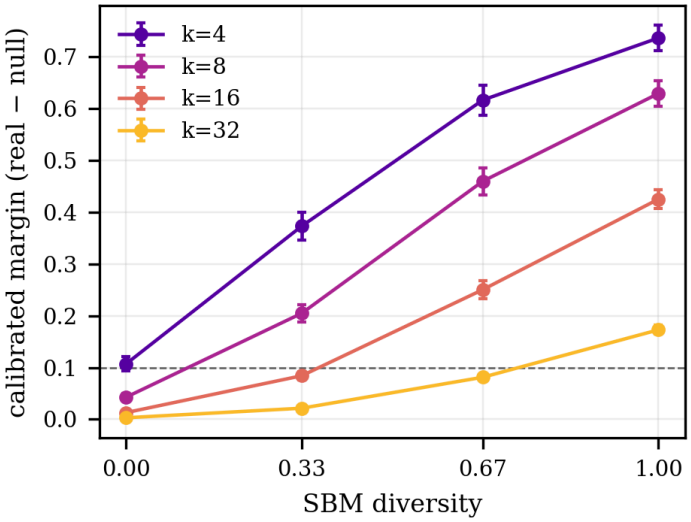
Four diversity levels $d \in \{0.0, 0.33, 0.67, 1.0\}$ (rows) $\times$ four independent samples (columns), drawn under the Phase 4 parametrisation: $n = 50$ nodes, 4 blocks, $p_{\text{intra}} \in (0.3, 0.9)$, $p_{\text{inter}} \in (0.01, 0.2)$. At $d = 0$ every graph is identically distributed (midpoint of each range); at $d = 1$ each graph samples its own $(p_{\text{intra}}, p_{\text{inter}})$ independently from the full range. Per-graph edge counts annotated above each tile.



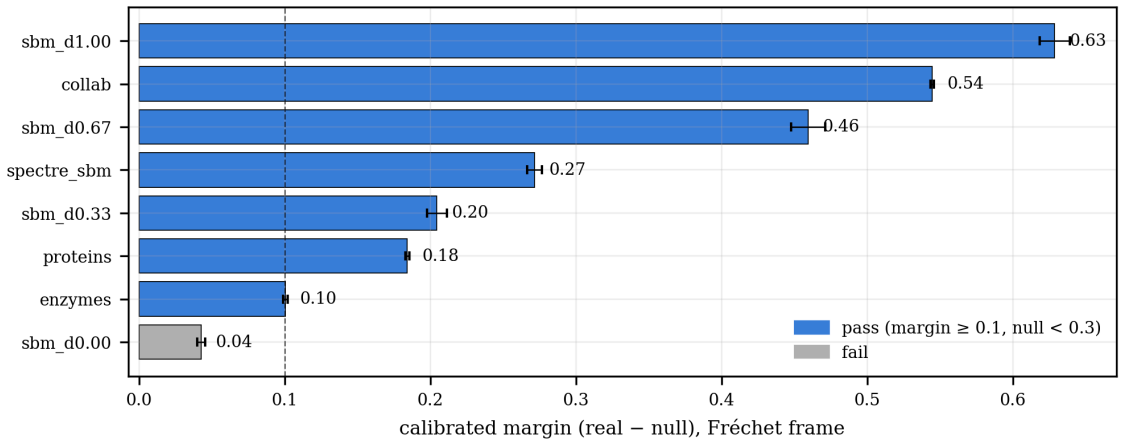
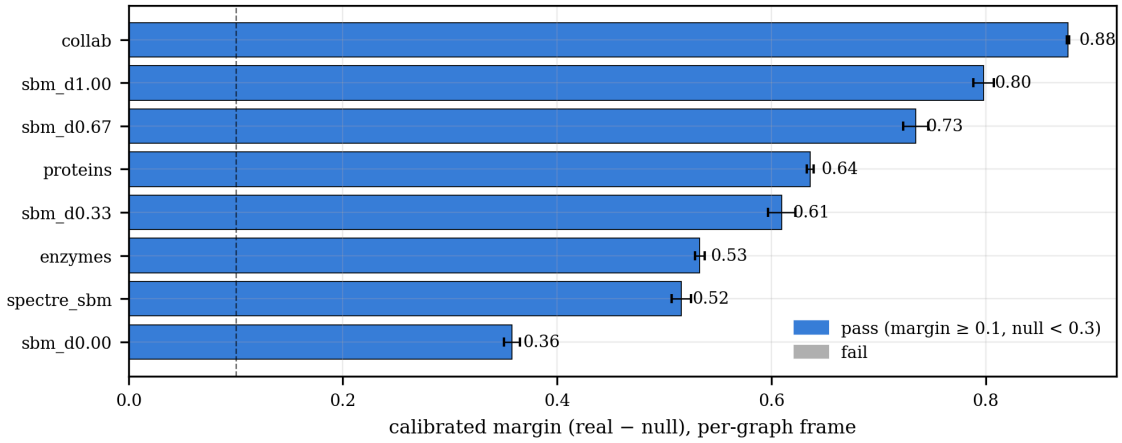
F1 — FVE vs ε, Gaussian (top) and DiGress (bottom)



F2 — calibrated margin vs SBM diversity



F3 — dataset ranking (Fréchet frame, headline cell)



FS1 — dataset ranking (per-graph frame)